

MINHAL RAHMAN

B.Tech Graduate – Applied Electronics & Instrumentation Engineering

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Kerala, India

PROFILE

Instrumentation and Automation Engineering graduate with exposure to industrial automation, PLC programming, SCADA, HMI, and process control through academic projects and industrial training. Currently pursuing a Professional Diploma in Industrial Automation at IPCS Calicut, with hands-on learning in PLC-based automation and industrial control systems. Interested in automation, control systems, and project engineering, with strong problem-solving skills and a willingness to learn and contribute in dynamic engineering environments.

TECHNICAL SKILLS

Automation & Control:

PLC Programming, SCADA, HMI, Process Control, Industrial Instrumentation

Embedded & Hardware

ESP32, Microcontroller Programming, Sensors & Actuators

PLC & Automation Tools:

RSLogix 500, Zelio Soft, CODESYS, LabVIEW, Godot OIP

Programming & Tools

C, MATLAB, VS Code

PROJECTS

Smart Liquid Mixer (ESP32 Based)

- Developed an ESP32-based automated liquid mixing prototype for process control applications.
- Built a web-based monitoring dashboard using HTML, CSS, and JavaScript for real-time visualization of system parameters.

PLC-Based Warehouse Automation System – CODESYS & Godot OIP

- Developed a virtual warehouse automation system with PLC-based conveyor and sorting control.
- Implemented Ladder Logic, Structured Text, and OPC UA communication for real-time control and monitoring.

LabVIEW Simulation of Digital Twin

- Designed an HMI front panel for real-time monitoring of process level and temperature parameters.
- Developed core process simulation logic with data visualization for monitoring system behavior and operating conditions.

WORK EXPERIENCE

SMEC Labs, Kozhikode

Industrial Automation Intern

May – June, 2025

- Gained exposure to PLC programming, SCADA systems, and industrial instrumentation.
- Worked with sensors, actuators, and control panels used in industrial automation systems.

EDUCATION

Government Engineering College, Kozhikode

B.Tech in Applied Electronics & Instrumentation Engineering | 2022–2026

CGPA: 7.24 / 10.00

IPCS Global, Calicut

Professional Diploma in Industrial Automation (PDIA) | 2026–Present

PLC · SCADA · HMI · Industrial Automation · Industrial Communication

CERTIFICATIONS & WORKSHOPS

- Siemens Mobility – Commercial Project Manager Job Simulation | Forage, 2026
Project planning, KPI analysis, EAC and stakeholder communication
- NPTEL: Control Engineering – IIT Madras
- Workshop: Embedded Systems & Automotive (ENDORA'23, GEC Kozhikode)